

CoAP Concept Study Report

Introduction:

CoAP (Constrained Application Protocol) is a lightweight application protocol designed for IoT devices. It runs over UDP and supports REST-like methods such as GET, POST, and PUT. It is suitable for low-power devices like ESP8266 sensors.

CoAP vs MQTT Comparison

Feature	CoAP	MQTT
Transport	UDP	TCP
Communication	Request/Response	Publish/Subscribe
Broker Needed	No	Yes
Power Usage	Lower	Higher
Best For	Battery sensors	Continuous telemetry

Resource Mapping Example

GET /sensors/temp returns current DHT22 temperature value.

Response: {temperature: 29.4}

Why CoAP for 55-second Sleep Cycle?

CoAP is ideal because the device can wake up, send a small UDP packet, receive acknowledgment, and return to deep sleep quickly. This reduces power consumption compared to MQTT, which requires TCP connection setup and broker communication.

Conclusion

Zelbytes would choose CoAP for battery-powered remote sensor clusters where devices wake briefly, transmit sensor readings, and return to deep sleep. CoAP improves power efficiency and reduces communication overhead.